Lezione 14 Metodi Numerici

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0.1 Stabilità assoluta dei metodi RK a s-stadi

$\begin{cases} \dot{y} = \lambda y \\ y(0) = 1 \end{cases} \Rightarrow u_{n=1} = R(z)u_n \quad R(z) = (1 + zb^T \sum_{j=0}^{s-1} (zA)^j \mathbb{1})$

RK2 s= 2 $\begin{cases} b_1 + b_2 = 1 \\ c_2 b_2 = \frac{1}{2} \end{cases} \Rightarrow \left(\begin{array}{c|cc} 0 & 0 & 0 \\ \frac{1}{2\alpha} & \frac{1}{2\alpha} & 0 \\ \hline & 1-\alpha & \alpha \end{array}\right)$

$R(z) = (1 + z(1-\alpha, \alpha)(I + zA)\mathbb{1})$

$$A = \begin{pmatrix} 0 & 0 \\ \frac{1}{2\alpha} & 0 \end{pmatrix} \quad b^T = (1-\alpha, \alpha), \quad (zA)^0 = I \quad (zA)^1 = zA$$

$I + zA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + z \begin{pmatrix} 0 & 0 \\ \frac{1}{2\alpha} & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{z}{2\alpha} & 1 \end{pmatrix}$

$(I + zA)\mathbb{1} = \begin{pmatrix} 1 & 0 \\ \frac{z}{2\alpha} & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{z}{2\alpha} + 1 \end{pmatrix}$

$$\begin{aligned} R(z) &= 1 + z(1-\alpha, \alpha) \begin{pmatrix} 1 \\ \frac{z}{2\alpha} + 1 \end{pmatrix} = 1 + z(1 - \alpha + \alpha(\frac{z}{2\alpha} + 1)) \\ &= 1 + z(1 - \cancel{\alpha} + \frac{z}{2} + \cancel{\alpha}) = 1 + z + \frac{z^2}{2} \quad \text{(Taylor al secondo ordine)} \end{aligned}.$$

Ricordiamo che dato il problema test:
$y(t) = e^{\lambda t} = e^{Re(\lambda)t} e^{iIm(\lambda)t}$
$y(t_{n+1}) = e^{t_{n+1}} = e^{\lambda(t_n + h)} = e^{\lambda t_n} \cdot e^{\lambda h} = y(t_n)e^z$
$y(t_{n+1}) = y(t_n)e^z = \ldots = u(0)(e^z)^n = (e^z)^n$

0.2 Pendolo (non lineare)

$$\begin{cases} x = l \sin\theta \\ y = -l\cos\theta \end{cases} \xrightarrow{\frac{d}{dt}} \begin{cases} \dot{x} = l\cos\theta\dot{\theta} \\ \dot{y} = l\sin\theta\dot{\theta} \end{cases}.$$

$T = \frac{1}{2}mv^2 = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2}ml^2\dot{\theta}^2$
$U = mgy + mgl = mgl(1 - \cos\theta)$
$L = T - U = \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1 - \cos\theta)$
$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$ Equazioni del moto
$\frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta} \to \frac{d}{dt}\frac{\partial L}{\partial \theta} = ml^2\ddot{\theta}$
$\frac{\partial L}{\partial \theta} = -mgl\sin\theta$
$\Rightarrow ml^2\ddot{\theta} + mgl\sin\theta = 0 \Rightarrow \ddot{\theta} + \frac{g}{l}\sin\theta = 0$
se $\theta \sim 0 \Rightarrow \sin\theta \sim \theta \Rightarrow \ddot{\theta} + \frac{g}{l}\theta = 0 \Rightarrow \ddot{\theta} + \omega^2\theta = 0$
$g = l = m = 1$

$$\begin{cases} \ddot{\theta} = -\sin\theta \\ \theta(0) = \theta_0 \\ \dot{\theta}(0) = \varphi_0 \end{cases} \Rightarrow \ddot{\theta} = -\theta.$$

$\dot{\theta} = \varphi$

$\dot{\varphi} = -\theta \Rightarrow \begin{pmatrix} \dot{\theta} \\ \varphi \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \theta \\ \varphi \end{pmatrix}$

$\begin{cases} \dot{\theta} = \varphi \\ \phi = 0 \sin\theta \end{cases}$

$Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \theta \\ \varphi \end{pmatrix} \quad F(Y) = \begin{pmatrix} y_2 \\ -\sin y_1 \end{pmatrix}$

$\dot{Y} = F(Y)$

$Y(0) = \begin{pmatrix} \theta_0 \\ \varphi_0 \end{pmatrix} = \begin{pmatrix} y_{10} \\ y_{20} \end{pmatrix}$

Forward Euler

$U^{n+1} = U^n + hF(U^n)$

Backward Euler

$U^{n+1} = U^n + hF(U^{n+1})$

$\mathcal{F}(x) = x - hF(x) - U^n$ cerco x^* tale che $\mathcal{F}(x^*) = 0$ Uso il metodo di punto fisso con newton

$$x^{(k+1)} = x^{(k)} - J_{\mathcal{F}}(x^{(k)})^{-1}\mathcal{F}(x^{(k)}).$$

$\begin{cases} J_{\mathcal{F}}(x^{(k)})(x^{(k+1)} - x^{(k)}) = -F(x^{(k)}) \\ x^0 = U^n \end{cases}$

Che riscriviamo come

$\begin{cases} J_{\mathcal{F}}(x^{(k)})W = -\mathcal{F}(x^{(k)}) \\ x^{(0)} = U^n \end{cases} \quad \to x^{(k+1)} = x^{(k)} + W$

si itera finché $\|x^{(k_1)} - x^{(k)}\| = \|W\| < tol$

$\mathcal{F}(x) = x - hF(x) - U^n$

$J_{\mathcal{F}}(x) = I - hJ_F(x)$

$J_{\mathcal{F}}(x) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - h \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}$

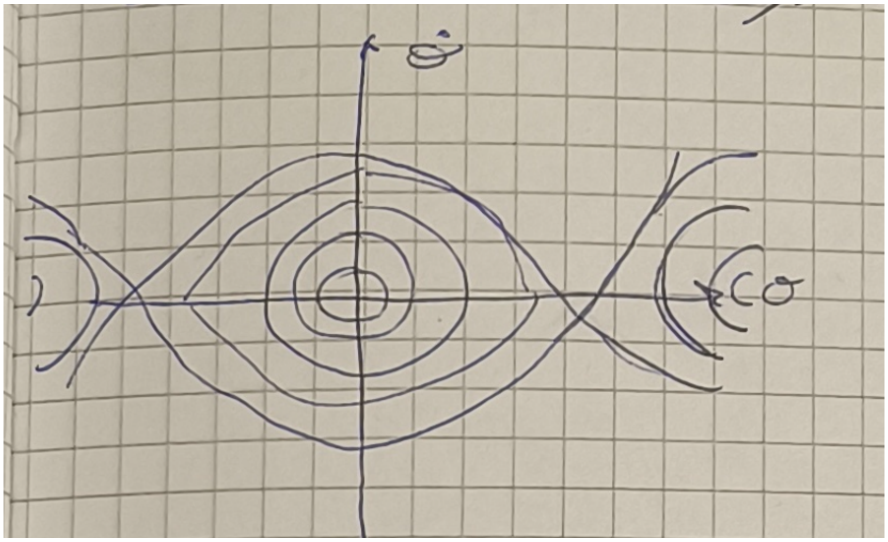
$J_F(x) = \begin{pmatrix} 1 & -h \\ h\cos x & 1 \end{pmatrix}$

Dove abbiamo usato $F(x) = \begin{pmatrix} x_2 \\ -\sin x_1 \end{pmatrix}$ e $J_F(x) = \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}$

$$\begin{cases} \begin{pmatrix} 1 & -h \\ h\cos x_1^{(k)} & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = -(x^{(k)} - hF(x^{(k)}) - U^n) \\ x_1^{(k+1)} = x_1^{(k)} + w_1 \quad x_1^{(0)} = u_1^n \\ x_2^{(k+1)} = x_2^{(k)} + w_2 \quad x_2^{(0)} = u_2^n \end{cases}.$$

Per vedere che il solver funzioni guardiamo l'energia

$E = T + U = \frac{1}{2}\dot{\theta}^2 + (1 - \cos\theta) \overset{\theta \sim 0}{\sim} \frac{1}{2}\dot{\theta}^2 + \frac{1}{2}\theta^2$ energia dell'oscillatore armonico



$\frac{d}{dt}E = \frac{d}{dt}\left(\frac{1}{2}\dot{\theta}^2 + 1 - \cos\theta\right) = \dot{\theta}\ddot{\theta} + \sin\theta\dot{\theta} = \dot{\theta}(\ddot{\theta} + \sin\theta) = 0$
$\Rightarrow E = cost = E(0) = \frac{1}{2}(\varphi_0)^2 + 1 - \cos\theta_0$